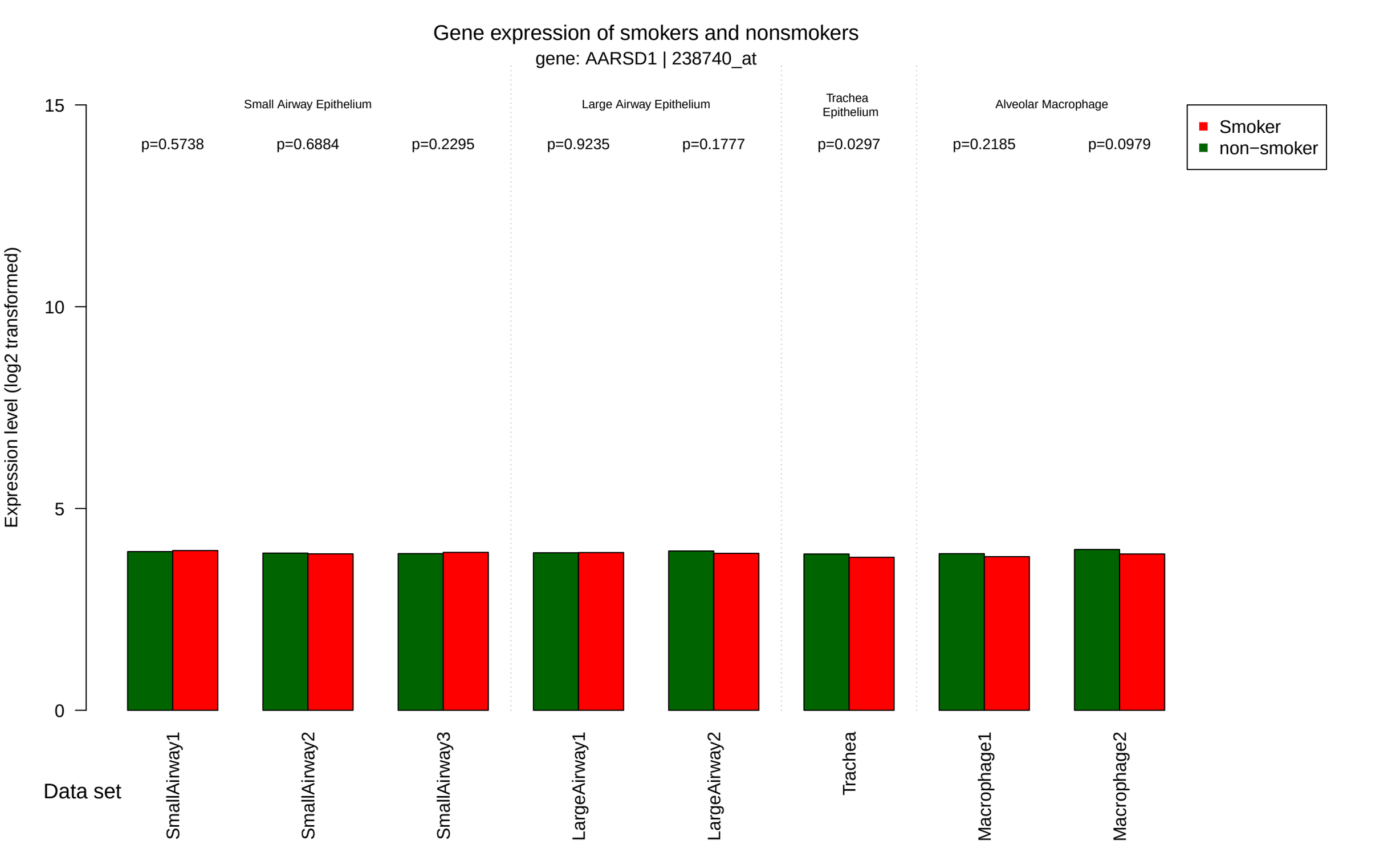
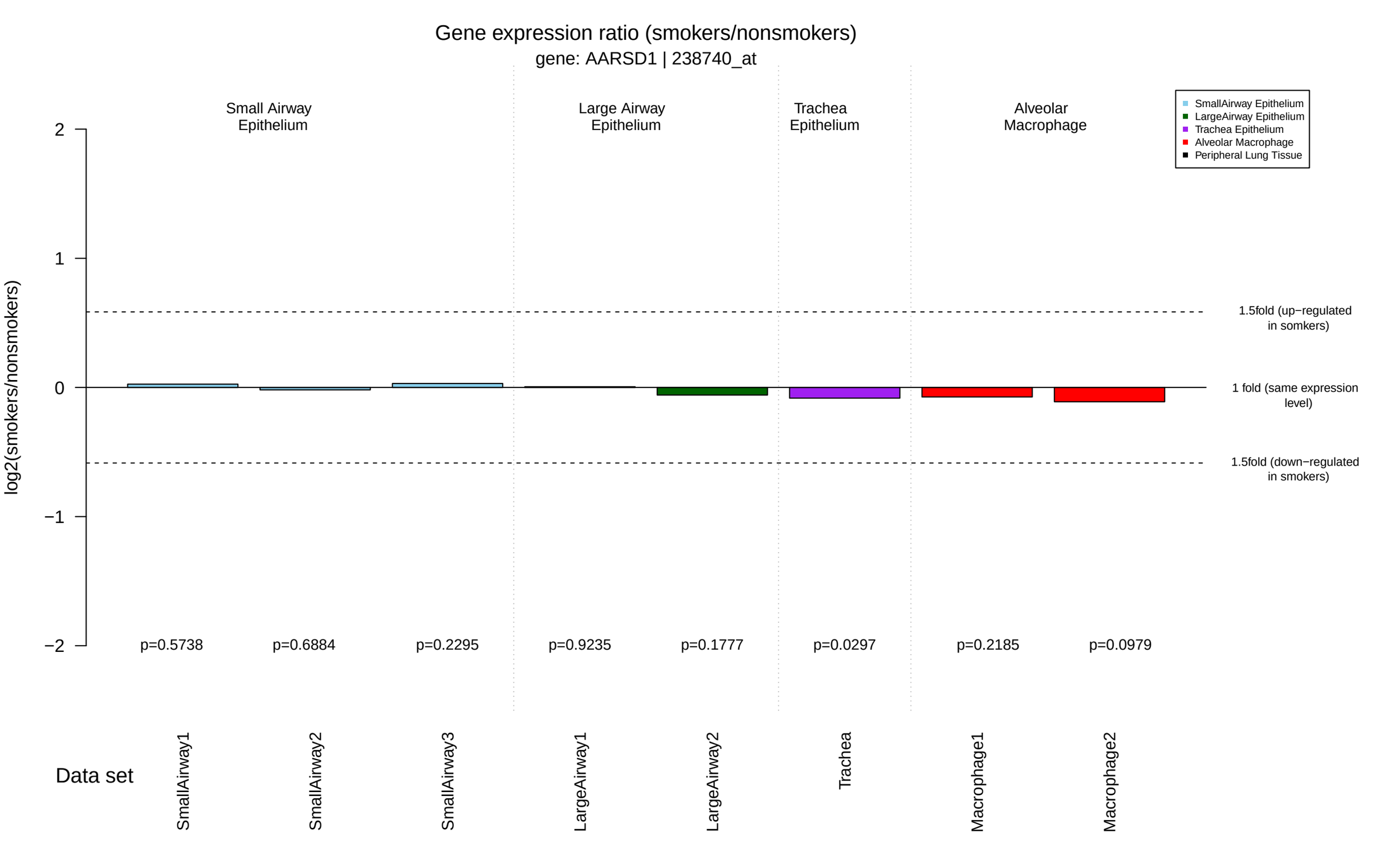
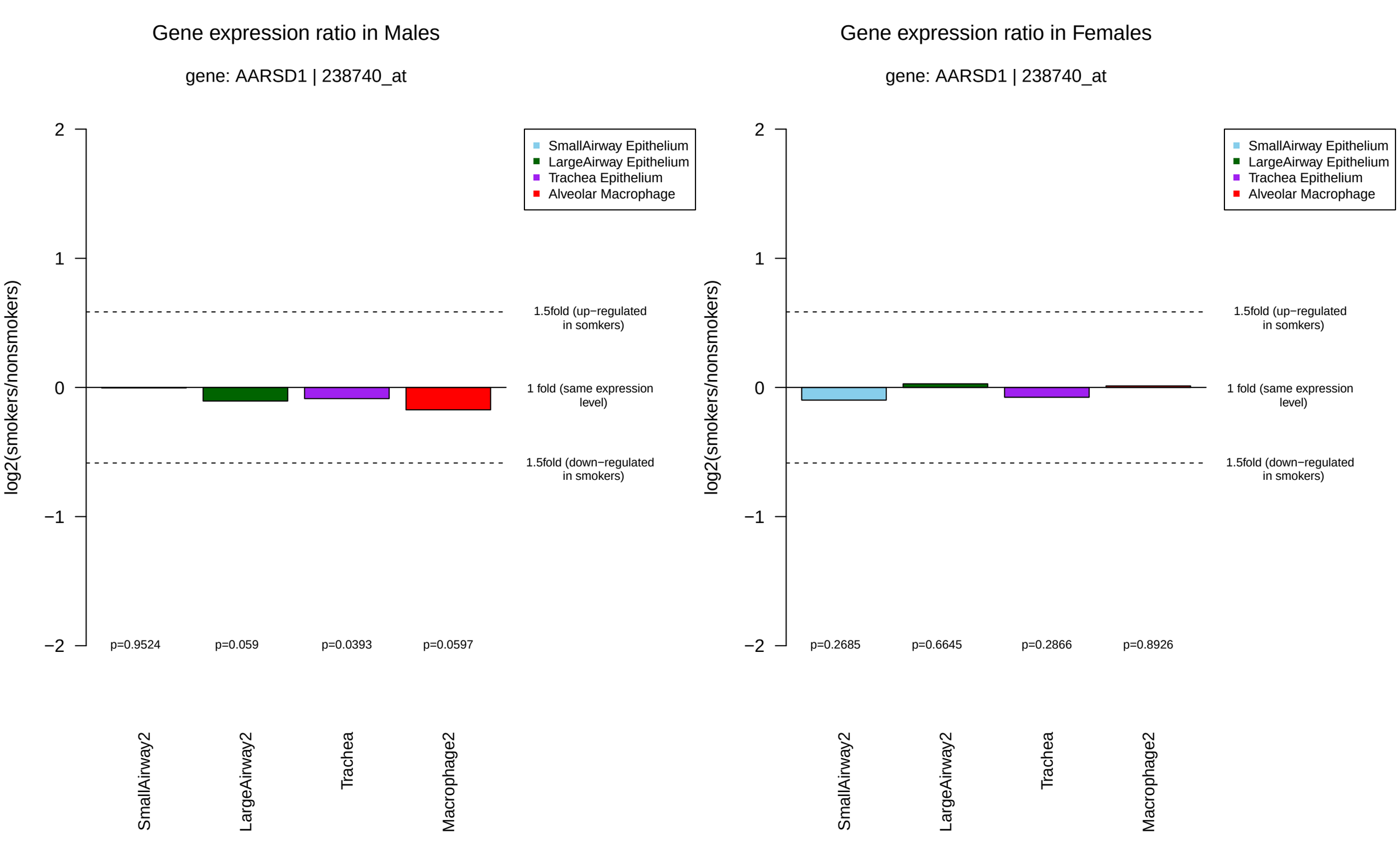
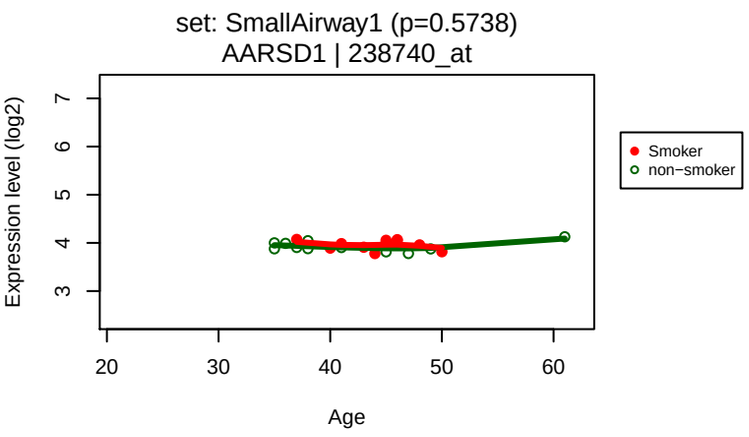
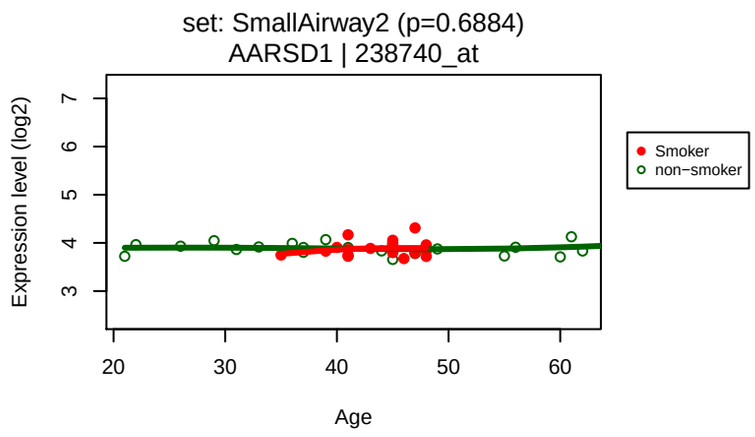
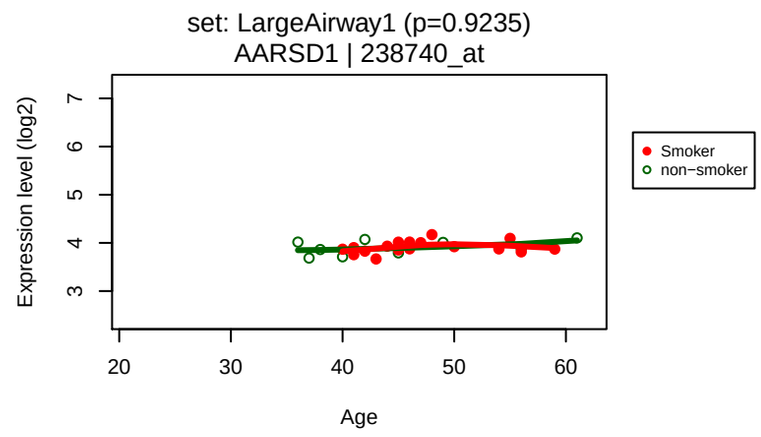

Gene expression of smokers and nonsmokers
gene: AARSD1 | 238740_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5738
p=0.6884
p=0.2295
p=0.9235
p=0.1777
p=0.0297
p=0.2185
p=0.0979
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

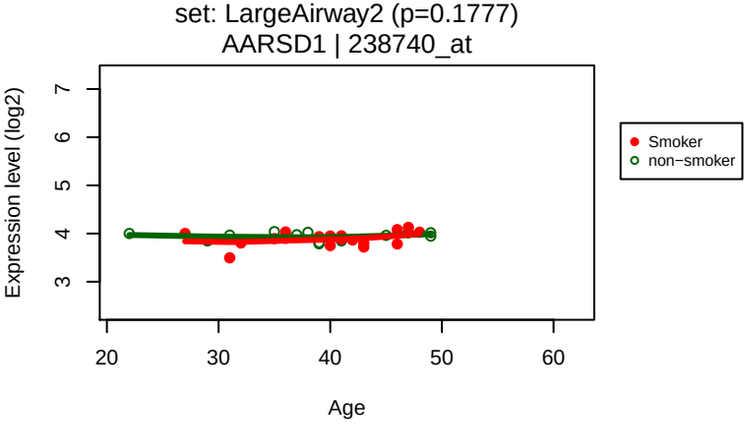

Gene expression ratio (smokers/nonsmokers)
gene: AARSD1 | 238740_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5738
p=0.6884
p=0.2295
p=0.9235
p=0.1777
p=0.0297
p=0.2185
p=0.0979
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

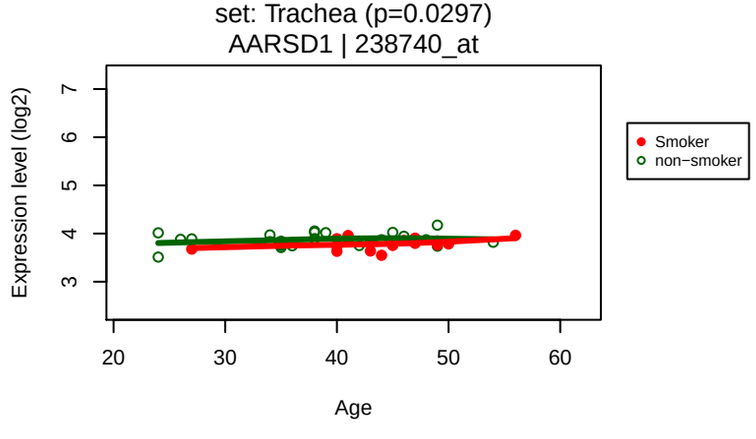

Gene expression ratio in Males
gene: AARSD1 | 238740_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9524
p=0.059
p=0.0393
p=0.0597
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AARSD1 | 238740_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2685
p=0.6645
p=0.2866
p=0.8926
SmallAirway2
LargeAirway2
Trachea
Macrophage2

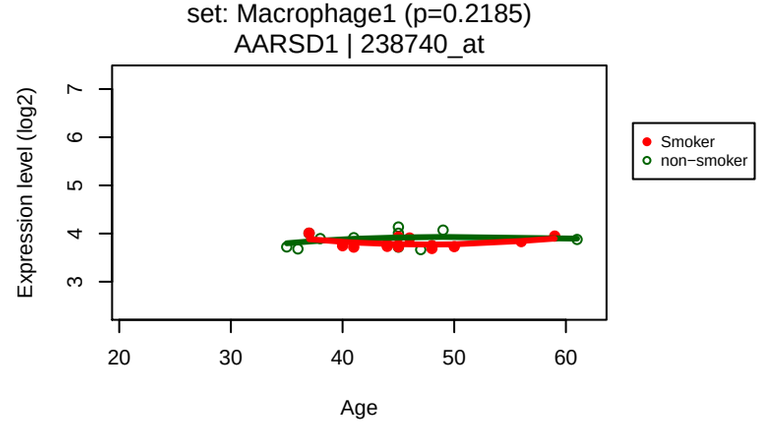

set: SmallAirway1 (p=0.5738)
AARSD1 | 238740_at
Expression level (log2)
Age
Smoker
non−smoker

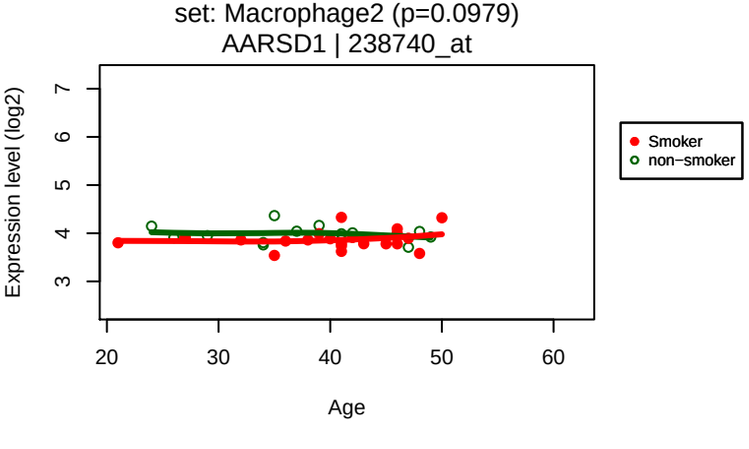

set: SmallAirway2 (p=0.6884)
AARSD1 | 238740_at
Expression level (log2)
Age
Smoker
non−smoker

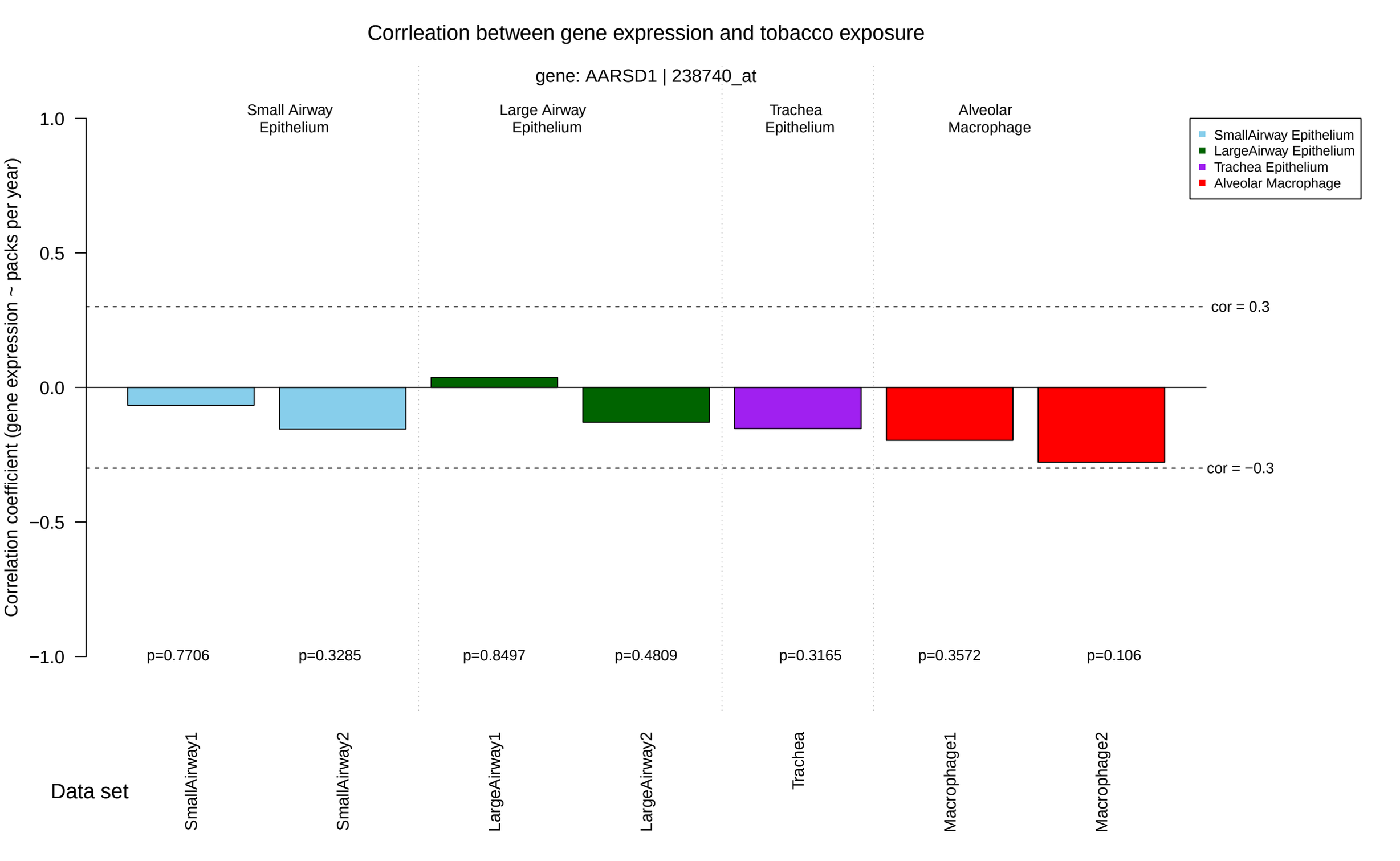

set: LargeAirway1 (p=0.9235)
AARSD1 | 238740_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1777)
AARSD1 | 238740_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0297)
AARSD1 | 238740_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2185)
AARSD1 | 238740_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0979)
AARSD1 | 238740_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AARSD1 | 238740_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7706
p=0.3285
p=0.8497
p=0.4809
p=0.3165
p=0.3572
p=0.106
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2